

DANIEL QUIGLEY

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STATEMENT

A mathematical semanticist studying the formal structure of intelligence: developing formal accounts of meaning, inference, grounding, and abstraction in natural and artificial systems. I prove theorems connecting formal semantics and logical reasoning to the distributional representations of neural models, formulate grounding as a measurable property, develop a relational calculus for abstraction and analogy, and analyze the geometry of semantic space.

EDUCATION

University of Wisconsin-Milwaukee

Milwaukee, WI | 2025

PhD: Linguistics

- Interdisciplinary PhD research: linguistics | mathematics | computer science | philosophy
- **Advisors:** Nicholas Fleisher, Hamid Ouali (linguistics) | Jeb Willenbring (mathematics) | Matthew Knachel (philosophy and logic)
- **Dissertation:** *Neurosymbolic Semantics*

University of Wisconsin-Milwaukee

Milwaukee, WI | 2023

MA: Linguistics

- Interdisciplinary MA research: linguistics | mathematics | computer science | philosophy
- **Advisors:** Nicholas Fleisher, Hamid Ouali (linguistics) | Susan McRoy (computer science)

Universiteit Utrecht

Utrecht, The Netherlands | 2019

MSc Certificate: Theoretical Physics

- Interdisciplinary Graduate Honors recipient
- Completed Master's coursework in Theoretical Physics and Mathematics; presented independent research on geography and climate science in the interdisciplinary honors seminar

University of Wisconsin-Madison

Madison, WI | 2018

BSc: Anthropology | Astronomy | Linguistics | Mathematics | Physics

- **Advisors:** Stefan Westerhoff (physics, astronomy, mathematics) | Monica Macaulay (linguistics) | J. Mark Kenoyer (anthropology)

RESEARCH EXPERIENCE

Indiana University Bloomington

Bloomington, IN | Oct 2025 – Present

Postdoctoral Researcher

- **Principal Investigator:** Jacob Gates Foster
- Extending a research program in vector-space semantics in which extensional, intensional, counterfactual, and hyperintensional formal semantics embed homomorphically into distributional models (extensional case: *Journal of Logic, Language and Information*, 2025; intensional case under review; counterfactual case in preparation), with results on modal completeness, logical omniscience failure, and recovery of classical logics as limiting cases.
- Formulated symbol grounding as a measurable, audited profile over declared evaluation standards and executed an end-to-end pilot audit of a pretrained language model (with E. Maynard; under review, *Minds and Machines*).
- Derived a longitude-resolved theory of the mean tropical year (arXiv, 2026); quantified discrepancies between stated and effective seasonal targets of historical calendar reforms and constructed provably drift-bounded time-varying leap rules (two manuscripts: one under review, one in preparation).
- Developing matched-null statistical methodology for the geometric analysis of high-dimensional embedding spaces and a spectral framework unifying principal component analysis, persistent homology, and diffusion geometry through the persistent Hodge Laplacian (two manuscripts in preparation).
- Developing a formal theory of abstraction selection in analogical reasoning: a single-sorted monoidal calculus of relations with first-order expressive power and a description-length and transfer criterion over candidate relational abstractions (coauthored manuscript in preparation).

University of Wisconsin-Milwaukee

Milwaukee, WI | Aug 2020 – May 2025

Graduate Researcher: Linguistics (MA 2023, PhD 2025)

- Defined a structured typology distinguishing mathematical linguistics, computational linguistics, and NLP, clarifying their scientific goals and interdisciplinary coherence.
- Developed a categorical framework that unifies formal semantic models, ensuring modular, order-independent processing.
- Proved structure-preserving homomorphisms between formal semantics, both extensional and intensional, and distributional vector space semantics.
- Designed a generalized vector logic that integrates typed compositionality within distributional semantics for logical reasoning tasks.

- Developed a taxonomy of grounding criteria for generalized grounding problems, including a resolution for the symbol grounding problem for neurosymbolic AI.
- Designed and proved PSPACE-hard algorithms for language processing based on extensional and intensional semantics.
- Derived tensor forms for semantic representations of various linguistic phrasal types and constructions in hyperbolic spaces.

University of Wisconsin-Madison

Madison, WI | Oct 2015 – May 2018

Indus Valley Civilization Undergraduate Researcher

- Reconstructed broken strings of written data from the Indus Valley Script using n -gram Markov chains and conditional entropy, and applied statistical analysis techniques to analyze the data using Python.
- Collaborated with international colleagues to analyze the Indus Valley Script data, resulting in the creation of sign frequency scores that showed the context in which different symbols were used.
- Presented results at an international academic conference.

University of Wisconsin-Madison

Madison, WI | Oct 2014 – May 2018

Wisconsin Baldwin Idea Grant Project Assistant

- Coordinated with team of students and academic advisors to work with Menominee elders with Menominee language data elicitation and transcription.
- Recorded and documented language data for preservation and revitalization, and prepared teaching materials for language preservation and revitalization efforts.
- Results of work integrated into Menominee reference grammar (to be published).

Wisconsin IceCube Particle Astrophysics Center

Madison, WI | Oct 2014 – May 2018

High Energy Astroparticle Physics Research Assistant

- Designed and implemented simulations, data acquisition systems, and visualizations for HAWC (High-Altitude Water Cherenkov) gamma-ray detector.
- Resolved discrepancies in gamma-ray results across four international experiments; wrote GPS data system in C++.
- Communicated results of simulations and technical developments with international teams, demonstrating strong collaboration and communication skills.

FELLOWS EXPERIENCE

NAM (Natural and Artificial Minds)

Princeton, New Jersey | Mar 2026

Diverse Intelligences Fellow

DISI (Diverse Intelligences Summer Institute)

St. Andrews, Scotland | Jul 2025

Fellow

IPAM (Institute for Pure and Applied Mathematics)

Los Angeles, CA | Sep 2024 – Dec 2024

Mathematics of Intelligences Visiting Researcher

INTERNSHIP EXPERIENCE

Fujitsu Limited

Sendai, Japan | Jun 2024 – Aug 2024

Visiting Scientist

- Collaborated on an 8-week project with Graduate-level Research in Industrial Projects for Students (G-RIPS), UCLA Institute for Pure and Applied Mathematics (IPAM) and Fujitsu Limited, focusing on explanatory and interpretable AI in causal modeling.
- Reverse-engineered underlying mathematics of Fujitsu's "Wide Learning" classification machine learning model for causal AI.
- Developed a logical framework for feature relationships in Pearlean causal models.
- Developed a classifier model for causal graph structures using graph-theoretic hierarchical metrics.
- Designed an interactive, accessible interface for causal graph representation, incorporating language, vision-based, and machine-readable features for explanatory and interpretable AI in causal modeling.

Apple

Cupertino, CA | Jul 2023 – Nov 2023

Career Experience: Production Engineer

- Developed, tested, and deployed demo content over-the-air and operating system scripting to channel stores across 250,000 iOS, tvOS, watchOS, and macOS devices from development to production.
- Developed and documented sophisticated automation frameworks using Python and Plist scripting to enhance operational efficiency.
- Resolved failing Wi-Fi connectivity across demo devices by investigating plist data structures for discrepancies; resolved content publishing issues by region and device.
- Validated content in twenty-one languages across twenty-five locales sensitive to local content and language requirements while crafting comprehensive test plans and technical documentation for new features and internal tools.

PRESENTATIONS, PUBLICATIONS, PREPRINTS

Presentations

- **Quigley, Daniel** (2026). *Updates since then: vector logic; analogies and abstractions; page margins*. UCLA Institute for Pure and Applied Mathematics.
- **Quigley, Daniel** (2026). *What's in a month?*. Bloomington, Indiana.
- **Quigley, Daniel** (2026). *Towards analogy and abstraction via strings*. Santa Fe Institute.
- **Quigley, Daniel** (2026). *Some (coarse) geometric and topological features of semantic space: geometric invariants of word embeddings and treeish model of semantic space*. Indiana University Bloomington.
- **Quigley, Daniel** (2025). *On vector logic for extensional formal semantics*. Logic Seminar. Indiana University Bloomington.
- **Quigley, Daniel** (2025). *Symbol grounding and other grounding problems*. Indiana University Bloomington.
- **Quigley, Daniel** (2025). *Basics of Python (for linguists)*. UW-Milwaukee.
- Joachim, Audrey and **Quigley, Daniel** (2025). *Interdisciplinarity*. Linguistics Society of America.
- **Quigley, Daniel** and Weller, Jae (2025). *Introducing SLAC: towards inter-university community*. Linguistics Society of America.
- **Quigley, Daniel** (2024). *Semantics of imperatives and imperative constructions*. Institute for Pure and Applied Mathematics Seminar.
- **Quigley, Daniel** (2024). *Geometric page design*. UCLA Institute for Pure and Applied Mathematics Seminar.
- **Quigley, Daniel** (2024). *Syntax and Semantics of Large Language Models*. Syntax and Semantics Reading Group.
- **Quigley, Daniel** (2024). *Semantics as Neurosymbolic*. UCLA Invited Speaker.
- **Quigley, Daniel** (2024). *On Accessibility: Screen-accessibility and PDF design*. Institute for Pure and Applied Mathematics Seminar.
- Forde, John and Mendez, Gaspar and Okubo, Akane and **Quigley, Daniel** and Sakamoto, Renji (2024). *Fujitsu Causal Discovery: a novel interactive platform for conditional causal discovery*. Fujitsu Limited.
- **Quigley, Daniel** (2024). *Be Reasonable! Relating Logical Models and Vector Spaces for NLP Interpretability*. Workshop in General Linguistics.
- **Quigley, Daniel** (2024). *Getting Started with $\text{\LaTeX}$* . Workshop in General Linguistics.
- **Quigley, Daniel** (2024). *Merge: Syntax-Semantics as a Hopf Algebra*. Algebraic Structures Seminar.
- **Quigley, Daniel** (2024). *Natural Language Understanding as Tensor Product Models*. Algebraic Structures Seminar.
- **Quigley, Daniel** (2024). *A Primer on the Mathematics of Artificial Neural Networks*. Graduate Student Colloquium.
- **Quigley, Daniel** (2023). *Tensor Space and Category-Theoretic Semantics for Resolving Long-Distance Linguistic Expressions in Natural Language Processing*. PhD preliminary paper and presentation.
- **Quigley, Daniel** (2023). *Decoding Authorial Style, Tone, and Mood in Poetic Translations through Natural Language Processing: An Analysis of Beowulf*. Workshop in General Linguistics.
- **Quigley, Daniel** (2023). *$\text{\LaTeX}$ for Linguists*. Summer Workshop.

Publications

- **Quigley, Daniel** (2026). *page-canons: Classical page-layout canons with margin-note support*. CTAN.
- **Quigley, Daniel** (2025). *A vector logic for extensional formal semantics*. Journal of Logic, Language and Information. arXiv preprint, 2024.
- **Quigley, Daniel** (2025). *Neurosymbolic Semantics*. PhD dissertation. University of Wisconsin-Milwaukee.
- **Quigley, Daniel** et al. (2024). *White Paper: Mathematics of Intelligences*. IPAM Long Program.
- Forde, John and Mendez, Gaspar and Okubo, Akane and **Quigley, Daniel** and Sakamoto, Renji (2024). *Fujitsu Causal Discovery: a novel interactive platform for conditional causal discovery*. Fujitsu Limited.
- **Quigley, Daniel** (2024). *Be Reasonable! Relating Logical Models and Vector Spaces for NLP Interpretability*. In Proceedings: Workshop in General Linguistics.
- **Quigley, Daniel** (2023). *Decoding Authorial Style, Tone, and Mood in Poetic Translations through Natural Language Processing: An Analysis of Beowulf*. In Proceedings: Workshop in General Linguistics.

Under review

- Maynard, Eric and **Quigley, Daniel**. *On measuring grounding and generalizing grounding problems*. Under review: Minds and Machines. arXiv preprint, 2025.
- **Quigley, Daniel**. *A vector logic for intensional formal semantics*. Under review: Journal of Logic, Language and Information. arXiv preprint, 2026.
- **Quigley, Daniel**. *An analytic framework for historical page canons*. Under review: Journal of Mathematics and the Arts.
- **Quigley, Daniel**. *A Social and Structural Definition of a Calendar*. Under review: KronoScope.

In preparation

- Beckelhymer, Daniela and **Quigley, Daniel** and Flores, Cynthia and Ortega, Omayra. *From Language to Landfills: Mathematics, Ethics, and the Governance of AI*. Manuscript in preparation.
- **Quigley, Daniel**. *Mean-year arithmetic and seasonal anchors in calendar reform*. Manuscript in preparation.
- **Quigley, Daniel**. *A vector logic for counterfactuals*. Manuscript in preparation.

- Craimer, Avi and Haydon, Nathan and **Quigley, Daniel**. *A relational calculus on a pairing-closed domain*. Manuscript in preparation.
- **Quigley, Daniel**. *Word embedding geometry under matched-null testing*. Manuscript in preparation.
- **Quigley, Daniel**. *The Hodge Laplacian under principal component analysis, persistent homology, and diffusion geometry*. Manuscript in preparation.

Preprints

- **Quigley, Daniel** (2026). *Mean tropical year length at arbitrary ecliptic longitude*. arXiv.
- **Quigley, Daniel** (2026). *Formal specification and behavioral simulation of the holiday gift exchange game*. arXiv.
- **Quigley, Daniel** (2024). *Categorical Framework for Typed Extensional and Intensional Models in Formal Semantics*. arXiv.

PROJECT EXPERIENCE

Indiana University Bloomington

Bloomington, IN | Oct 2025 – Present

LaTeX Development

- Reverse-engineered and derived algorithms for page layout and design according to traditional and contemporary design principles and developed *LaTeX* packages for modular control over text area and margin layout.
- Developed AA machine accessible custom lettrine package for decorative initials and in-line letter decoration.
- Developed accessible graphic user interface for designing, visualizing, and implementing text area and margin layout, with *LaTeX*-compatible output and general typesetting parameters for use outside of *LaTeX*.

University of Wisconsin-Milwaukee

Milwaukee, WI | Aug 2020 – May 2025

LaTeX Development

- Designed *LaTeX* document templates, accepted by university as official resources for graduate school.
- Created document tagging and readability methods to improve designs of accessible PDF documents.
- Developing intelligent UIs for improved accessibility of PDF documents, improving usability for users with accessibility needs and machine readability.

University of Wisconsin-Milwaukee

Milwaukee, WI | May 2021

Data Visualization and Sentiment Analysis of Movie Reviews across Four Neural Network Models

- Conducted a comprehensive comparison of performance metrics for polarity sentiment analysis of movie reviews using deep learning techniques, including the design of four different machine learning architectures (CNN, RNN, RCNN, LSTM).
- Compared the performance of the different model architectures across ten epochs, with a cutoff for validation loss, and achieved an accuracy rate of greater than 83% for each model.
- Demonstrated expertise in the application of deep learning techniques to natural language processing tasks through the successful execution and analysis of this research project.

University of Wisconsin-Milwaukee

Milwaukee, WI | May 2021

ML Optimization: No Free Lunch

- Optimized, evaluated, and compared performance scores for classification machine learning tasks: Decision Tree Classifier; K-Nearest Neighbor; Multinomial Naive Bayes; Logistic Regression; SVC; Dummy Classifier; Neural Network.
- Optimized, evaluated, and compared performance scores for regression machine learning tasks: Decision Tree Regressor; Linear Regression; SVR; Dummy Regressor; Neural Network.
- Evaluated CNN architectures of image classification task using the Fashion-MNIST dataset.

University of Wisconsin-Milwaukee

Milwaukee, WI | Jul 2020

Linux from Scratch

- Completed *Linux from Scratch* project, building a fully functional Linux distribution from scratch using source code and following project documentation, demonstrating strong problem-solving and troubleshooting skills to resolve issues during build process.
- Developed deep understanding of Linux operating system, including kernel, system libraries, and userland utilities, and improved skills in working with source code and building software from ground up.
- Created a customized Linux distribution that met specific needs and preferences, showcasing ability to tailor a system to meet unique requirements.

WORK EXPERIENCE

VersAI

Milwaukee, WI | Jan 2025 – Present

Principal Researcher

- Designed formally specified framework for LLM-based requirements verification: three-valued verdict semantics (Kleene lattice with defined collapse policies); ten evaluation primitives; ISO/IEC 25023-aligned KPIs (FRIC, VFRC, ACC, UOC) covering implementation, verification depth, and objective satisfaction.
- Developed statistical calibration layer: soundness estimation with Clopper-Pearson intervals; labeling-noise correction; importance-reweighted operational estimands gated by Kish effective sample size, and per-stratum drift detection across releases.
- Authored delivery program: six-phase plan with written exit gates; nine scoped external work packages; gold-set annotation protocol with inter-annotator agreement; governance specification for the framework's calibration hyperparameters.

Obviate

Milwaukee, WI | Nov 2025 – Present

Contract: Machine Learning Scientist

- Audited the operator algebra of a typed risk propagation engine for LLM-driven contract analysis; identified a sign defect on conflict edges and two further semantic errors, and derived corrected per-type operators from legal conflict typologies.
- Restored contraction without spectral normalization and designed an LLM-gated propagation architecture with per-branch convergence guarantees.
- Delivered a 44-page technical report, an investor diligence brief, and a pre-registered validation protocol with fixed falsification conditions.

University of Wisconsin-Milwaukee

Milwaukee, WI | Aug 2024 – May 2025

Research Assistant

- Designed regression models and engineered relevant features for predicting battery health in production in collaboration with Clarios, achieving 89% accuracy in predicting battery production performance.
- Developed algorithms for data explanation and interpretability of causal models and relationships for features of battery health, using LiNGAM causal models and classification.
- Developed visual representations for data, causality, and explainability for battery health and failure, presented to international team to be implemented in production.

Eruditis

Milwaukee, WI | Jun 2024 – Present

Contract: Machine Learning Scientist

- Developed mathematical models for AI-enhanced algorithmic trading systems intended for non-institutional investors.
- Derived mathematical model and representation for financial metrics adaptable to investor profile parameters.
- Active mentor for team development of financial analyses and mathematical content.

Apple

Glendale, WI | Oct 2021 – Mar 2024

Genius Technician

- Demonstrated leadership while also mentoring Technical Specialists and Technical Experts | developed and implemented new processes to improve efficiency and effectiveness of Genius Bar team.
- Exceeded expectations for customer satisfaction: attained performance review scores of 88 TMS and 74 NPS, excelling in metrics for technical expertise (89) and empathy (80).
- Certified for iPhone and Mac repair, maintaining 95% repair rate on devices.

University of Wisconsin-Milwaukee

Milwaukee, WI | Aug 2020 – May 2025

Instructor of Record

- Responsible for class sizes of 20-30 students per semester, providing comprehensive support and guidance.
- Designed course content to include topics in natural language processing, such as introductory concepts and artificial intelligence ethics.
- Providing effective feedback and communication to improve performance, demonstrating commitment to student success and learning.

GRADUATE TEACHING EXPERIENCE

Linguistics 100: Instructor of Record	Spring 2025
Linguistics 100: Instructor of Record	Fall 2024
Linguistics 100: Instructor of Record	Spring 2024
Linguistics 100: Instructor of Record	Fall 2023
Linguistics 210: Instructor of Record	Spring 2023
Linguistics 100: Instructor of Record	Fall 2022
Linguistics 210: Instructor of Record	Spring 2022
Linguistics 210: Instructor of Record	Fall 2021
Linguistics 210: Teaching Assistant	Spring 2021
Linguistics 210: Teaching Assistant	Fall 2020

GRADUATE COURSEWORK

Linguistics: Phonetics | Phonology | Morphology | Syntax | Semantics | 2nd Language Acquisition | Seminar: Ellipsis | Typology and Universals | Historical and Comparative Linguistics | Seminar: Research Methods | Seminar: Double Object Constructions | Advanced Phonetics | Advanced Phonology | Advanced Syntax | Advanced Semantics | Foundations of Formal Logic | Advanced Independent Study

Computer Science: Machine Learning and Applications | Introduction to Natural Language Processing | Introduction to Artificial Intelligence | Artificial Intelligence in Business | Advanced Independent Study | Advanced Machine Learning | Algorithm Design and Analysis

Physics: Quantum Field Theory | Statistical Field Theory | General Relativity | String Theory | Field Theory in Particle Physics | Cosmology | Radiative Processes | High Energy Astrophysics

Mathematics: Differential Geometry | Geometric Partial Differential Equations | Mathematical Methods in Theoretical Physics | Algebraic Structures

PROFESSIONAL INTERESTS

Linguistics: semantics | formal logic | semiotics | computational linguistics | mathematical models | model theory | ellipsis | double object constructions | transitivity mismatches | case stacking | construction grammar | scope | binding | degree and comparison | typology | language change | analogies

Computer Science: grounding problems | algorithm design | complexity | formal logic | mathematical models | model theory | machine learning methods for language processing | human language technologies | human-computer interaction

Artificial Intelligence: neural networks | natural language processing | explainable artificial intelligence | geometric neural networks | graph neural networks | semantic spaces | abstraction | analogies

Physics: gravitational physics | black hole physics | early universe physics | topological defects | quantum field theory in curved spacetime | inverse problem for Lagrangians

Mathematics: category theory | group theory | differential geometry | geometric PDEs | geometric flows | Ricci flow | operator theory | formal logic | model theory | inverse problems | abstraction

Anthropology: writing | calendrical systems | power and social relations | gender | ethnoarchaeology | archaeoastronomy

PROFESSIONAL AFFILIATIONS

American Mathematical Society (AMS)

American Physical Society (APS)

Association for Computational Linguistics (ACL)

Association for the Advancement of Artificial Intelligence (AAAI)

Language Creation Society (LCS)

Linguistic Society of America (LSA)

HONORS AND AWARDS

University of Wisconsin-Milwaukee: <i>Graduate Teaching Assistantship</i>	Aug 2020 – May 2025
University of Wisconsin-Milwaukee: <i>Chancellor's Graduate Student Award</i>	2025
University of Wisconsin-Milwaukee: <i>Chancellor's Graduate Student Award</i>	2023
University of Wisconsin-Milwaukee: <i>Chancellor's Graduate Student Award</i>	2020
Universiteit Utrecht: <i>Graduate Honors</i>	2019
University of Wisconsin-Madison: <i>Record - Number of Majors (5)</i>	2018

SKILLS

Data skills: Technical writing | Data collection, annotation, processing, visualization, statistical analysis, machine learning (Python: NumPy, Keras, Scikit-Learn, NLTK, Pandas, Matplotlib, TensorFlow, Mathematica, LiNGAM) | Technical documentation

Project skills: Written and oral presentation and communication | Qualitative and quantitative research methods | Problem solving | Experiment design | Language data collection and analysis | Team work and team leadership

Technical skills: Python | $\text{\LaTeX}$ | Excel | Praat | R | SPSS | regexp | SQL (basic) | HTML (basic) | Jekyll (basic)

Operating systems and software: Linux | Windows | MacOS | Conda | CUDA (GPU Programming) | MS Office Suite

LANGUAGES

Native: English

Elementary: German | Dutch | Finnish | Japanese

Some study: Menominee | Arabic (MSA) | Sanskrit | Georgian